DON BOSCO
INSTITUTE OF TECHNOLOGY
Bengaluru, Karnataka

AWS STUDENT
BUILDER GROUP
DBIT

AFTERHOURS 1.0

CODE BEYOND THE CLOCK

THEME 01: AGENTIC AI & MULTI-AGENT SYSTEMS

APPLYCOPILOT

Voice-Steered Autonomous Web Agent with 5 Specialized Execution Protocols

"Speak your intent. Watch Jarvis navigate, reason, and execute — across transit rushes, civic portals, fraud defense, and anti-dark-pattern checkouts."

TEAM DETAILS

Team Name: AI Avengers

Track: Theme 01: Agentic AI & Multi-Agent Systems

Don Bosco Institute of Technology (DBIT)

TEAM MEMBERS

- **Raj Aanada**
- **Yashma C P**
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- **Yashwanth K R**

Problem Statement & Its Relevance

The Universal Web Friction & The 5 Critical Digital Failure Vectors. *From frantic 90-second ticket rushes to rejected 20KB documents and predatory dark-pattern fees.*

HIGH-STRESS RUSHES

TRANSIT PANIC & REPETITIVE FORMS

- **Transit Rush Panic:** Peak KSRTC and Tatkal train seats vanish in < 90 seconds due to slow manual field entry.
- **Multi-Portal Burnout:** Re-entering identical credentials, marks, & addresses across 20+ fragmented portals.
- **Bespoke Essay Exhaustion:** Drafting custom statements of purpose for scholarships under tight deadlines.

BOT BREAKDOWN

WHY EXISTING TOOLS FAIL

- **Voice Assistants Can't Click:** Siri and Alexa only answer questions; they cannot browse websites, select seats, or type.
- **Chrome Autofill Collapses:** Blind key-value dumps fail on dynamic seat grids, dates, and strict file upload rules.
- **Scrapers Get Banned:** Headless bots trigger Cloudflare 403 blocks and have zero safety bridge for CAPTCHAs or UPI.

< 90 SEC SELLOUT

Window before peak KSRTC/Tatkal seats sell out; manual typing fails under rush pressure.

38% REJECTION RATE

Government job & exam applicants blocked by rigid 20KB-50KB image upload limits.

HIDDEN TRAPS

REJECTED DOCS & SNEAKY FEES

- **Impossible Upload Constraints:** Portals demand photos strictly between 20KB-50KB and 3.5x4.5cm, causing rejection loops.
- **Deceptive Dark Patterns:** Sneaky pre-checked ₹199 insurance and auto-added convenience fees inflate totals.
- **Civic & Vernacular Exclusion:** Elderly and regional language citizens cannot navigate multi-step government SPAs.

EMPIRICAL SCALE

THE SCALE OF DIGITAL FRICTION

- **$100M+ Unclaimed Aid:** Scholarships, grants, and subsidies abandoned annually due to portal paperwork exhaustion.
- **< 2.5 Min Sell-out Window:** Peak transit reservations expire before manual form completion can finish.
- **₹1,200 Cr Dark-Pattern Drain:** Lost annually by consumers to deceptive pre-selected add-ons and hidden surge fees.

₹350+ SNEAKY FEES

Hidden insurance & convenience surcharges auto-checked during online ticketing checkouts.

CONVERSATIONAL EXECUTION FLOW

Voice Intent → Vault Match → Protocol Reason → HITL Gate → Exec

Speak your goal. Swarm navigates live, pausing for payment approval.

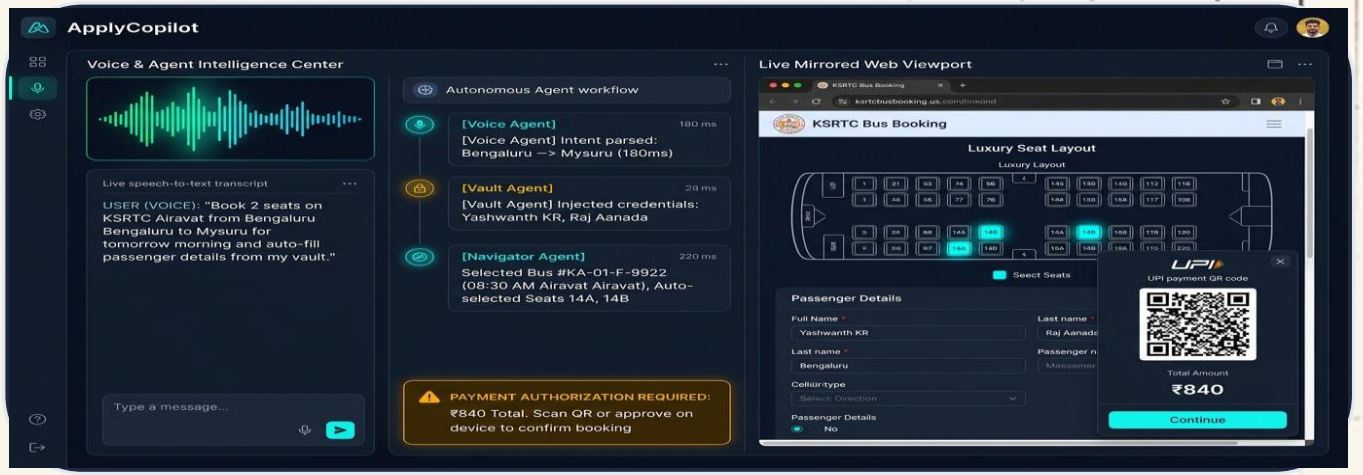
Proposed Solution & Core Innovation

ApplyCopilot: Autonomous Voice Engine Orchestrating 5 Core Protocols. *"Speak naturally. Jarvis dynamically engages the target protocol — from high-speed transit checkout to anti-dark-pattern sanitization."*

ARVIS WEB ARCHITECTURE

WHAT ARE WE PROPOSING?

- **Conversational Voice Steering:** Sub-250ms voice assistant transforms natural speech into structured browser actions.
- **5 Specialized Execution Protocols:** Dynamic task dispatch across transit booking, civic aid, document surgery, dark-patterns & fraud.
- **Air-Gapped Human Gate:** Mirrored viewport streams live clicks; auto-pauses for UPI payment & CAPTCHA clearance.



SPEED & SEATS

1. Transit Rush

- **Sub-250ms Grid Parser:** Instantly identifies vacant sleeper/window berths on KSRTC & IRCTC.
- **Passenger Vault Auto-Fill:** Injects names, age, and ID proofs to beat the 90-sec sellout window.
- **Direct-to-Checkout:** Bypasses promo pages straight to payment terminal.

CIVIC & SCHOLAR

2. Civic Helper

- **Vernacular Navigation:** Voice assistance in Kannada, Hindi & English for Seva Sindhu & Parivahan.
- **Grounded Essay RAG:** Synthesizes tailored scholarship motivation letters from stored achievements.
- **Elderly Inclusion:** Hands-free completion of complex multi-step government forms.

ZERO-TRUST SHIELD

3. Fraud Sentinel

- **Domain Provenance:** Audits registration age & SSL against phishing lookalikes (<30 days flag).
- **Air-Gapped Payment Gate:** State machine freezes DOM on checkout/UPI; never charges blindly.
- **Human-in-the-Loop:** Mandatory user authorization for UPI PIN, OTP, & submission.

IN-FLIGHT SURGERY

4. Document Exorcist

- **20KB-50KB Auto-Fixer:** Detects strict portal upload constraints in DOM validation attributes.
- **In-Memory Optimization:** Auto-crops to 3.5x4.5cm, sets 200 DPI, and compresses 2MB files to 38KB.
- **Zero Image Software:** Bypasses upload errors without touching MS Paint or Photoshop.

CONSUMER DEFENSE

5. Anti-Dark-Pattern

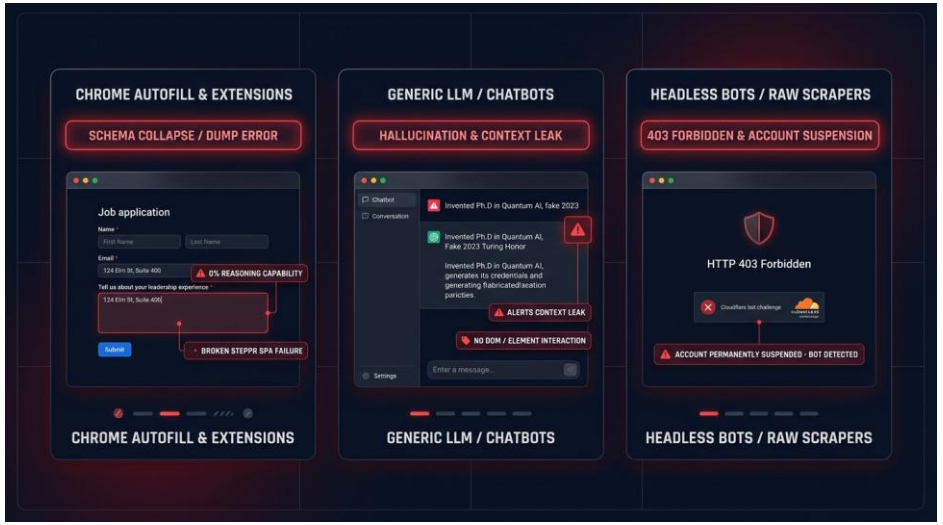
- **Hidden Fee Stripper:** Programmatically unchecks sneaky ₹199 insurance and donation add-ons.
- **Dynamic Pricing Guard:** Rotates incognito sessions & headers to neutralize cookie-based surge fees.
- **Clean Checkout Total:** Ensures users pay true minimum base fares across booking portals.

Existing Solutions & Differentiation

Why Voice Assistants and Autofill Extensions Fail on the Modern Web. *How ApplyCopilot's 5 protocols bridge conversational steering with real-world browser execution.*

LEGACY TOOL FAILURES

EXISTING TOOLS & CRITICAL BREAKDOWNS



- **Voice Assistants (Siri/Alexa): Conversational toys; zero DOM clicking or seat selection.**
- **Chrome Autofill: Rigid text dump; crashes on dynamic seat grids, calendars, & strict upload rules.**
- **Headless Bots: Trigger Cloudflare 403 blocks; zero handling for UPI/OTPs or dark-pattern fees.**

COMPETITIVE MATRIX

BENCHMARK: APPLYCOPILOT VS ALTERNATIVES

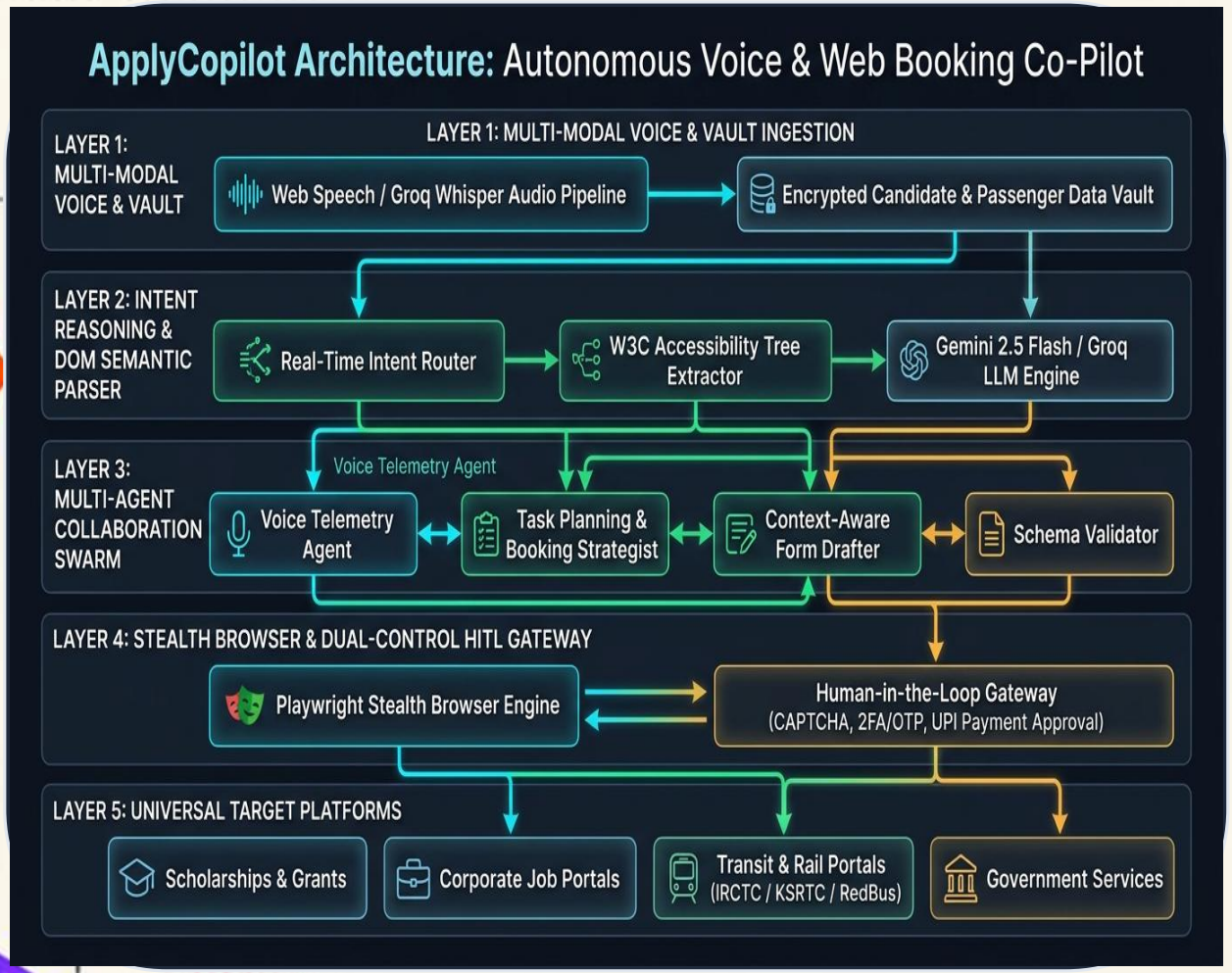
5 Core Capabilities	Siri / Alexa	Autofill	ChatGPT	ApplyCopilot
Transit Rush & Dynamic Seats	None	None	Blind	✓ Automated Grids
Civic Forms & Essay RAG	Blind	Rigid Dump	Hallucinates	✓ Grounded RAG
Document Exorcist (20-50KB)	None	Upload Error	None	✓ In-Flight Surgery
Anti-Dark-Pattern Fee Stripper	None	Pays Extra	None	✓ Auto-Unchecks Fees
Zero-Trust Fraud & UPI Gate	None	Blind Submit	None	✓ Air-Gapped UPI

OUR UNFAIR ADVANTAGE:

ApplyCopilot unites conversational voice AI with 5 specialized browser execution protocols. It solves high-stress booking rushes, tedious government forms, strict document limits, and predatory dark patterns while guaranteeing 100% human payment sovereignty.

Technical Approach & Feasibility

Voice Pipeline, Multi-Agent Swarm, Stealth Browser, and 24h Scope. *Sub-second conversational voice loop backed by Playwright and free-tier LLM inference.*



MULTI-AGENT SWARM

5 SPECIALIZED PROTOCOL WORKERS

- **Transit Worker:** Extracts web accessibility trees to parse dynamic seat grids & calendars.
- **Civic & Scholar Worker:** RAG embedding reasoner generating grounded motivation essays from vault.
- **Document Exorcist Worker:** In-memory Pillow/Wasm image resizer and PDF byte sanitizer (20KB-50KB).

STEALTH ENGINE & FRAUD GATE

STEALTH RUNTIME & AIR-GAPPED GATEWAY

- **Playwright-Stealth:** Masks navigator.webdriver, WebGL fingerprints, and browser headers against bot-detection.
- **Live Screencast Stream:** Streams CDP viewport frames via WebSockets directly to the web dashboard.
- **Payment Interrupt Gate:** State machine shifts to PAUSED on checkout/UPI/CAPTCHA; prompts user on mirrored canvas.

24-HOUR SCOPE

100% FREE DEVELOPER STACK & MVP

- **Zero Paid API Blockers:** Built on free Web Speech API, Groq Whisper, Gemini 2.5 Flash, Groq LPU, & Playwright.
- **Multi-Protocol 24h MVP:** Executes live transit seat selection, document auto-resizing, and fee stripping.
- **Failsafe Local Mock:** Local KSRTC/scholarship test server guarantees 100% demo uptime under spotty Wi-Fi.
- **Strict Human Sign-off:** Structured review diff ensures complete auditability before money or data is sent.

Expected Impact & Execution Plan

Eliminating Administrative & Transactional Friction via Voice Precision. *A concrete 24-hour sprint roadmap and measurable real-world outcomes.*

MEASURED VALUE

QUANTIFIABLE OUTCOMES

- **80% Faster Bookings:** Reduces transit ticket checkout from 8 mins to under 90 seconds (Transit Rush).
- **100% Document Pass Rate:** Zero upload rejections on strict 20KB-50KB exam/job portals (Document Exorcist).
- **₹350+ Saved Per Booking:** Immediate elimination of auto-checked insurance & surge fees (Anti-Dark-Pattern).
- **0% Payment Fraud:** Mandatory air-gapped human approval gate protects users from unintended charges.

TARGET BENCHMARKS

EVALUATION TARGETS

- **< 250ms Voice Intent:** Real-time speech transcription & structured command extraction via Groq.
- **> 96% Field & Seat Accuracy:** Correct identification of inputs, dates, and bus seat layouts.
- **< 1.5s In-Flight Document Resizing:** Sub-second image re-encoding to strict 20KB-50KB threshold.
- **100% Pre-Submit Audit:** Complete visual verification diff presented before final execution.

STAKEHOLDERS

WHO BENEFITS?

- **Daily Commuters & Travelers:** Instant hands-free bus/train ticket bookings during peak rush windows.
- **Students & Exam Applicants:** Painless scholarship submissions and auto-corrected document uploads.
- **Online Shoppers & Consumers:** Complete protection against deceptive dark-pattern up-sells.
- **Differently-Abled & Seniors:** Empowering voice-to-action co-pilot assistance on complex public portals.

24-HOUR SPRINT TIMELINE

RIGOROUS 24-HOUR HACKATHON EXECUTION ROADMAP

PHASE 1: 00:00 — 06:00
VOICE PIPELINE & VAULT SETUP

- Web Speech API & audio intent router setup.
- Encrypted local JSON vault for passenger/user credentials.
- Sub-250ms Groq Whisper streaming audio pipeline.

PHASE 2: 06:00 — 12:00
TRANSIT RUSH & CIVIC SWARM

- Playwright accessibility tree extractor for seat grids.
- KSRTC & scholarship portal interaction state machine.
- Grounded RAG for tailored motivation essays.

PHASE 3: 12:00 — 18:00
EXORCIST & ANTI-DARK-PATTERN

- In-memory image/PDF 20KB-50KB resizing engine.
- DOM selector rules unchecking hidden fees & insurance.
- Live CDP screencast stream to web dashboard.

PHASE 4: 18:00 — 24:00
FRAUD SENTINEL & LIVE DEMO

- Air-gapped interrupt loop on UPI/CAPTCHA/checkout.
- Side-by-side battle demo rehearsal for hackathon judges.
- Final latency optimization & pitch deck rehearsal.

Conclusion & References

The Conversational Web: Autonomous Execution With Absolute Human Sovereignty. *"Shifting from passive clicking to intelligent, voice-driven copilots that respect user control."*

CORE SUMMARY

KEY TAKEAWAYS

- **Voice-First Web Navigation:** Replaces exhausting manual clicking with instant natural speech for forms and bookings.
- **5 Specialized Protocols:** Executes Transit Rush, Civic Helper, Fraud Sentinel, Document Exorcist, and Anti-Dark-Pattern tasks.
- **Ethical Machine-Human Synergy:** Proves that autonomous agents must keep the human in full sovereign control over money and data.
- **Zero-Cost Production Stack:** Fully buildable in 24 hours using free Web Speech API, Gemini 2.5 Flash, Groq LPU, & Playwright.



TECHNICAL EVIDENCE

VERIFIABLE REFERENCES & SCIENTIFIC BENCHMARKS

1. **W3C Web Speech & ARIA Standards:** SpeechRecognition API & Accessibility Tree Navigation Specifications.
2. **Groq LPU Speech Benchmarks:** Sub-200ms Whisper Audio Transcription on Custom Inference Hardware.
3. **Google Gemini 2.5 Flash:** Low-Latency Multimodal Reasoning & Zero-Shot Schema Classification.
4. **Browser-Use & Stagehand (2024-2026):** Autonomous Web Agent Architecture & Natural Language DOM Routing.
5. **Playwright Stealth:** Mitigating Automation Fingerprints on High-Traffic Reservation Systems.
6. **FTC Guidelines on Automated Systems:** Mandatory Human Oversight & Consumer Consent in Digital Transactions.